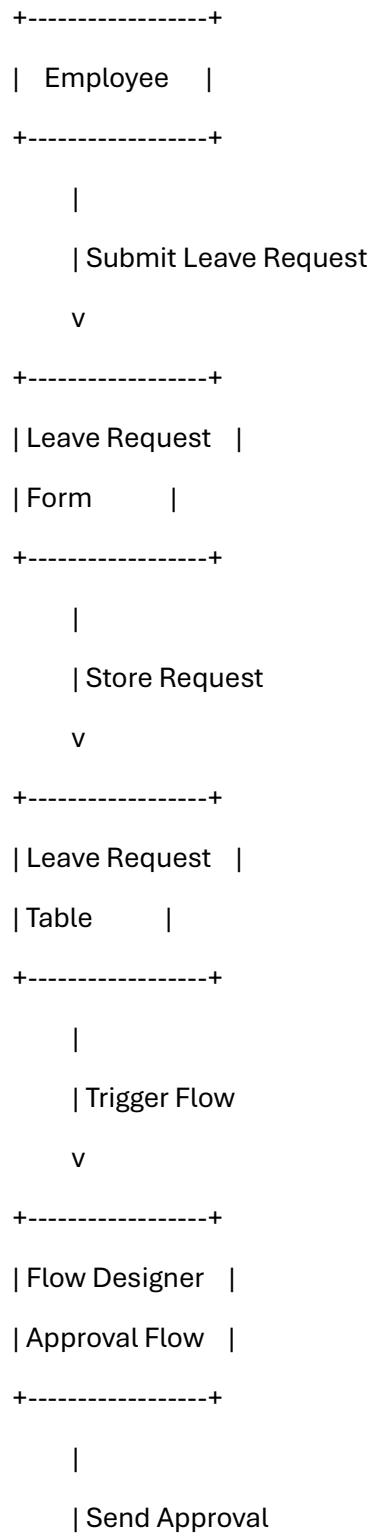
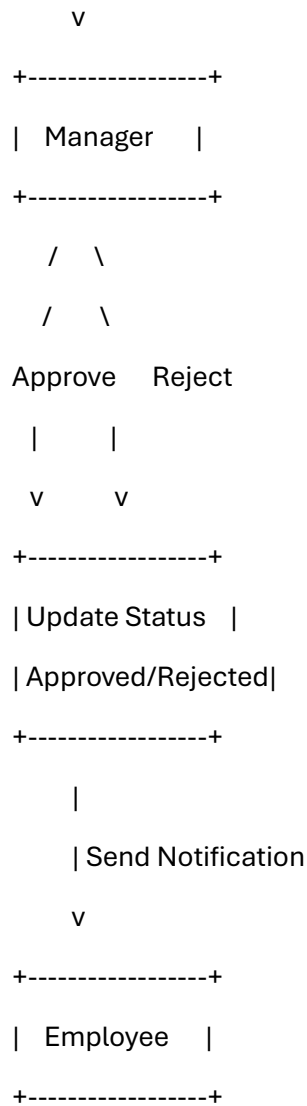


System Architecture

The Leave Approval System is designed using ServiceNow's workflow automation capabilities. The system enables employees to submit leave requests, which are automatically routed to managers for approval or rejection through Flow Designer. The application maintains leave records and updates the request status based on managerial decisions.

Architecture Flow





Components Used

Employee

- Submits leave requests.
- Views request status.
- Receives approval or rejection notifications.

Leave Request Form

- Captures leave details.
- Includes leave type, dates, reason, and manager.

Leave Request Table

- Stores all leave request records.
- Maintains status information.

Flow Designer

- Automates the approval process.
- Routes requests to managers.
- Updates status automatically.

Manager

- Reviews leave requests.
- Approves or rejects requests.

Notification Service

- Sends notifications regarding approval or rejection decisions.
-

Implementation Steps

Step 1: Login to ServiceNow Developer Instance

- Open ServiceNow Personal Developer Instance.
 - Login using administrator credentials.
-

Step 2: Set Global Scope

- Select **Global** from the Application Picker.
 - Ensure all development activities are performed under the Global scope.
-

Step 3: Create Update Set

Navigate to:

System Update Sets → Local Update Sets

Create a new update set named:

Leave Approval System

Click **Make Current**.

Step 4: Create Leave Request Table

Navigate to:

System Definition → Tables

Create a new table:

Property Value

Label Leave Request

Name u_leave_request

Save the table.

Step 5: Create Table Fields

Create the following fields:

Field Name	Data Type
Employee	Reference (User)
Manager	Reference (User)
Leave Type	Choice
Start Date	Date
End Date	Date
Reason	String
Status	Choice
Manager Comments	String

Step 6: Configure Choice Fields

Leave Type

Create choices:

- Casual Leave
- Sick Leave
- Earned Leave
- Emergency Leave
- Work From Home

Status

Create choices:

- Pending
- Approved

- Rejected

Set default value:

Pending

Step 7: Configure Form Layout

Arrange fields in the following order:

1. Employee
2. Manager
3. Leave Type
4. Start Date
5. End Date
6. Reason
7. Status
8. Manager Comments

Save the form layout.

Step 8: Create Application Menu

Navigate to:

System Applications → Studio

Create a new application:

Leave Approval System

Step 9: Create Application Modules

Create the following modules:

Leave Requests

Displays all leave requests.

My Leave Requests

Displays employee-specific leave requests.

Manager Approvals

Displays pending approval requests.

Step 10: Create Client Script for Date Validation

Navigate to:

System UI → Client Scripts

Create an **On Submit Client Script**.

Purpose:

- Prevent End Date from being earlier than Start Date.

Validation logic:

```
if(endDate < startDate){  
    return false;  
}
```

Step 11: Create Flow Designer Workflow

Navigate to:

Flow Designer

Create a flow named:

Leave Approval Flow

Step 12: Configure Flow Trigger

Add Trigger:

Record Created

Table:

Leave Request

Step 13: Configure Approval Action

Add action:

Ask For Approval

Configuration:

- Record = Leave Request Record
 - Table = Leave Request
 - Approver = Manager
-

Step 14: Configure Approval Logic

Approved Path

Condition:

Approval State = Approved

Actions:

- Update Status = Approved
- Send Approval Notification

Rejected Path

Condition:

Approval State = Rejected

Actions:

- Update Status = Rejected
 - Send Rejection Notification
-

Step 15: Configure Email Notifications

Approval Notification

Subject:

Leave Request Approved

Sent to Employee.

Rejection Notification

Subject:

Leave Request Rejected

Sent to Employee.

Step 16: Save and Activate Flow

- Save the flow.
- Click **Activate**.

Ensure the flow status is Active.

Step 17: Test Leave Request Submission

Create a new leave request.

Example:

- Employee = Fred Luddy
- Manager = Abel Tuter
- Leave Type = Sick Leave
- Reason = Fever

Submit the request.

Verify:

Status = Pending

Step 18: Test Manager Approval

Impersonate Manager.

Navigate to:

My Approvals

Open the request.

Select:

Approve

or

Reject

Save the record.

Step 19: Verify Results

Check:

- Status changes to Approved or Rejected.
 - Approval record is created.
 - Notification email is generated.
 - Leave request history is maintained.
-

Step 20: Project Completion

The Leave Approval System is successfully implemented with:

- Leave Request Management
- Date Validation

- Manager Approval Workflow
- Status Tracking
- Email Notifications
- Centralized Leave Records
- Automated Workflow using Flow Designer